High school · olympiad thinking

The pigeonhole principle

Progressive exercises

Exercise 1

Show that among 25 people, at least three were born in the same month.

Exercise 2

Choose six distinct numbers from $\{1,\ldots,10\}$. Prove that two sum to 11.

Exercise 3

Prove that among eight integers, two have a difference divisible by 7.

Exercise 4

Five points lie in a square of side 2, including its boundary. Show that two are at distance at most $\sqrt{2}$.